Evaluation of Equipment Models of Clustered Photolithography Tools for Fab-level Simulation

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Presentation Overview

- Motivation
- System Description: Clustered Photolithography Tool (CPT)
- Equipment Models
  - Linear model
  - Affine models
  - Flow line models (Improved)
- Numerical Experiments
  - Description (Three types of simulation)
  - Results
- Concluding Remarks

Motivation

Motivation (1) - CPT

- Clustered photolithography tools (CPT)
  - Cost up to $120 million[1], typically $20 – 50 million
  - Often the fabricator bottleneck
  - Key contributor to fab throughput capacity and cycle time

[2]

Motivation (2) – Fab-level Simulation

- High construction costs
  - Fabs must be well-designed and operated efficiently
- Fab-level Simulation
  - Essential decision support technology
  - Examples:
    - Detailed AMHS models (Jimenez et al. 2008, Hsieh et al. 2012)
    - Studies of fab behavior in relation to changes in lot size (Schmidt et al. 2006)
    - Cycle time reduction (Zarifoglu et al. 2008)
- Equipment models are key components of fab-level simulation

Motivation (3) – Features of Equipment Models

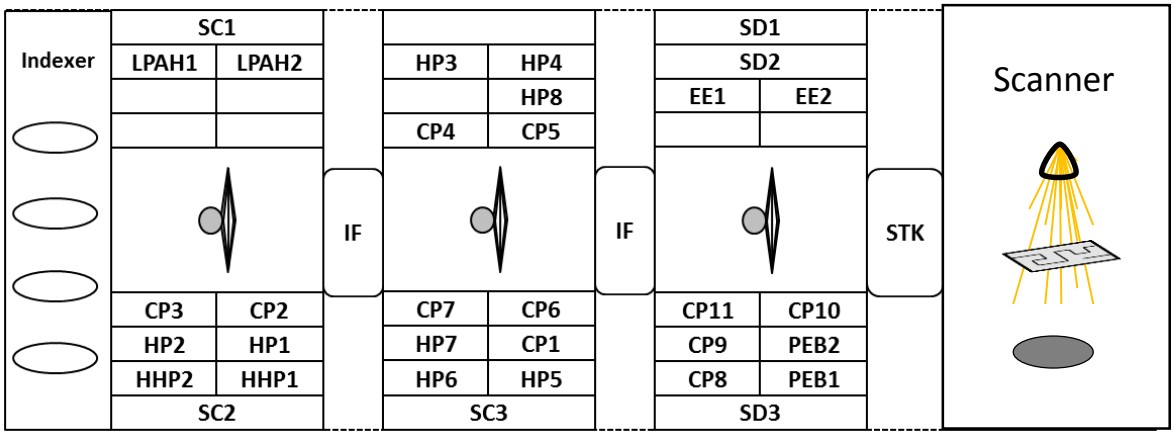
- Tradeoff: Fidelity vs. complexity
  - More detailed models lead to greater fidelity, but require longer computation times
  - Require more modeling effort
- Fab conditions can often change (different lot sizes, new toolsets, changing product mix, etc.)
  - Models often trained on specific set of input data, may not be robust when input conditions change

- **Goal:** Comparison of CPT Models for use in fab-level simulation
  - Accurate: Predict throughput with less than 1% error
  - Expressive: Incorporate fundamental behaviors
  - Computation: Very quick to calculate results
  - Robust: Less dependent on input data

System Description

System Description (1) – CPT

Clustered
Photolithography
Tool



- Multi-cluster tool, robot in each cluster, IF buffers, STK buffer
- Scanner is often the CPT bottleneck
- Largely deterministic process times
- Process time can vary by product
- Setups between lots (reticle changes, pre-scan setup, …)
- Wafer handling robot decision policy

System Description (2) – Performance Metrics

- **Notation**

  a_i : Arrival time of lot *i* to the tool

  S_i : Start time of lot *i* in the tool

  C_i : Completion time of lot *i* from the tool

  Lot class : $k_1 \in \{1, \ldots, .K\}$

  Number of wafers in lot *i* : W_i

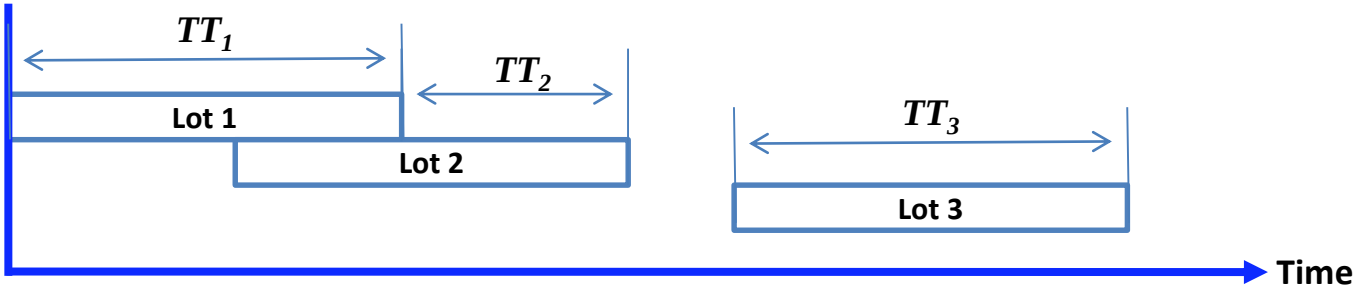
- **Performance measures**

  Cycle time of lot *i* : $CT_i = C_i - a_i$

  Lot residency time of lot *i* : $LRT_i = C_i - S_i$

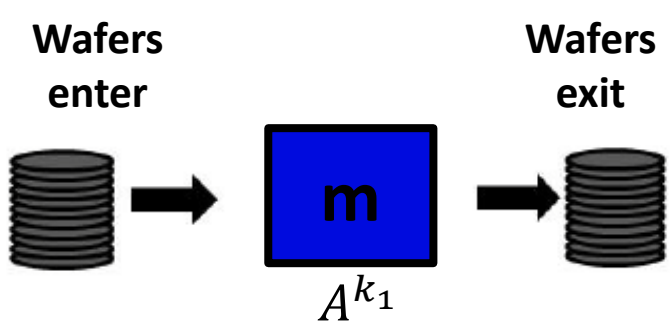
  Throughput time of lot *i* : $TT_i = min(C_i - S_i, C_i - C_{i-1})$

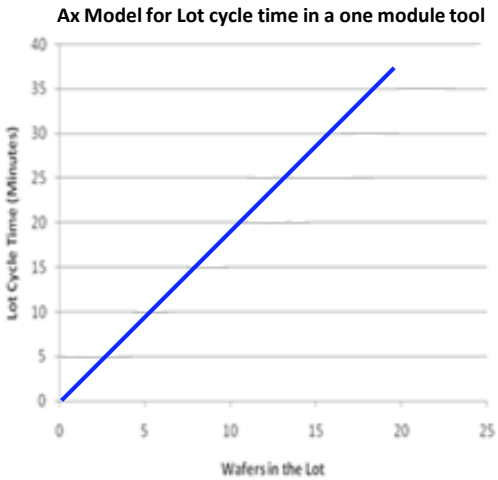
Computation time



Equipment Models

Equipment Model (1) – Linear Model





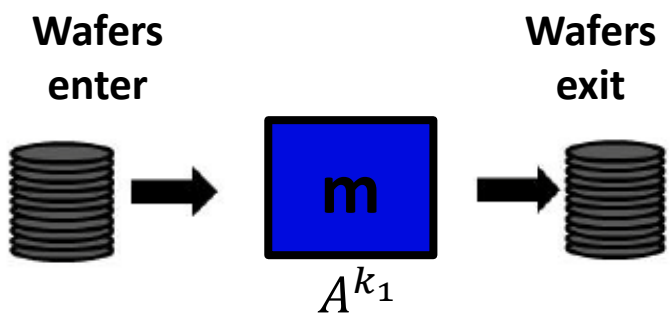
- Referred to as the Ax equipment model or linear model
  - Used to study recipe dedication in CPTs in an ASIC fab model[3]
- Lot indices per class : $L(k_1) = \{i | k(i) = k_1\}$
  - k_1 is current lot class
- Time between wafer completions: A^{k_1}
- Parameter estimation:

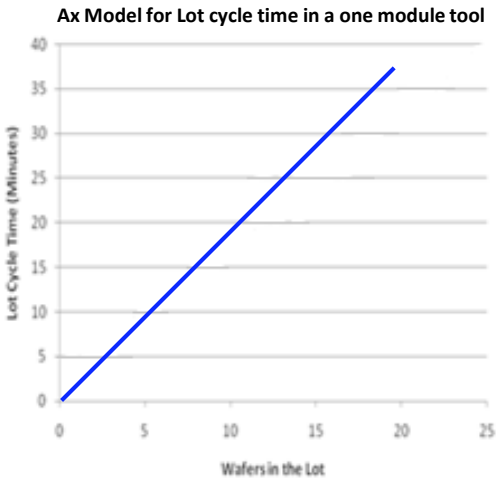
$$A^{k_1} = \frac{\sum_{i \in L(k_1)} (C_i - max(a_i, C_{i-1}))}{\sum_{i \in L(k_1)} W_i}$$

Complete Model:

$$\tilde{S}_i = \max\{a_i, \tilde{C}_{i-1}\}$$
$$\tilde{C}_i = \tilde{S}_i + A^{k_1} \times W_i$$

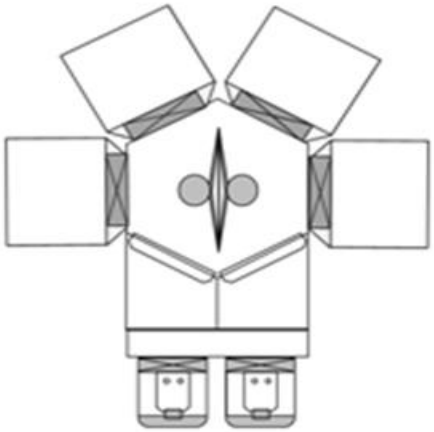
Equipment Model (1) – Linear Model

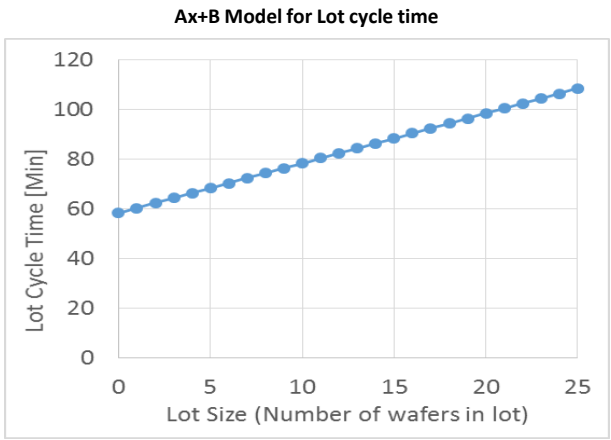




- Pros:
  - Simple to understand
  - Fast computation
- Cons:
  - Exactly matched to single module tool, not for CPT
  - New lots enter only when the tool is empty (No parallelism)

Equipment Model (2) – Affine Model





- Referred to as the Ax+B model
- Basic model provided in *AutoSched AP* [4]
- Lot indices per pairs of classes : $L(k_1, k_2) = \{i | k(i) = k_1, k(i-1) = k_2\}$
  - k_1 is current lot class, k_2 is previous lot class
- First wafer delay: B^{k_1,k_2}
- Time between wafer completions: A^{k_1}

$$A^{k_1} = \frac{\sum_{i \in L(k_1)} (C_i - max(a_i, C_{i-1}))}{\sum_{i \in L(k_1)} W_i}$$

$$B^{k_1,k_2} = \frac{1}{|L(k_1, k_2)|} \sum_{i \in L(k_1,k_2)} \left(C_{\Omega(i,1)} - \max(a_i, C_{i-1})\right)$$

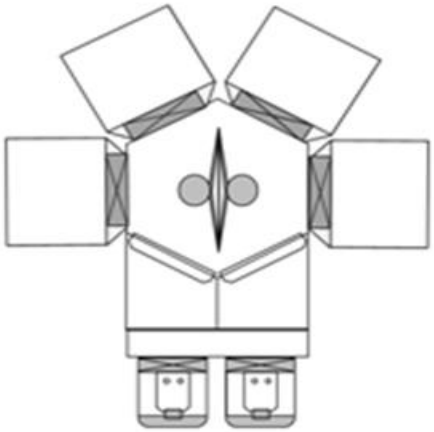
B is generalized to consider setups between classes

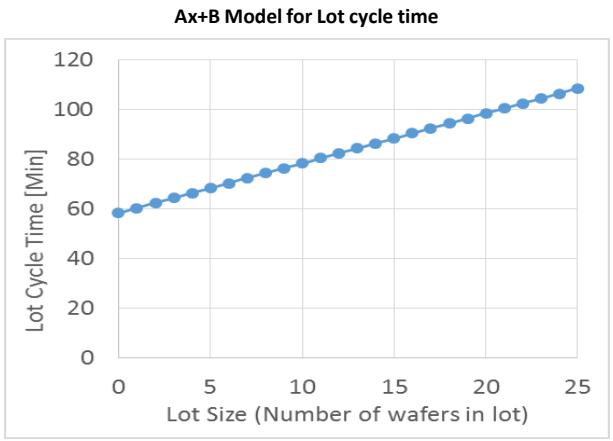
Complete model:

$$\tilde{S}_i = max(a_i, \tilde{C}_{i-1})$$

$$\tilde{C}_i = \tilde{S}_i + A^{k_1} \times (W_i - 1) + B^{k_1,k_2}$$

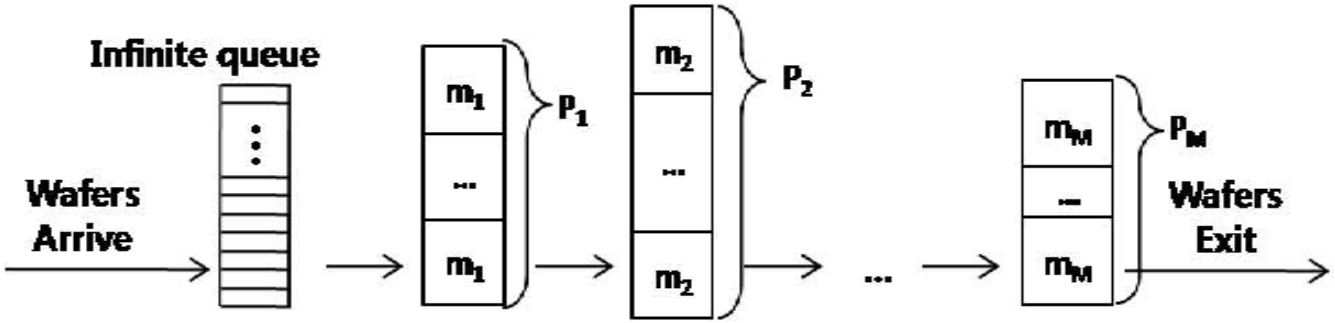
Equipment Model (2) – Affine Model





- Pros:
  - Simple to understand
  - Fast computation
- Cons:
  - Only one module per process, so not matched to CPT
  - New lots enter only when the tool is empty (No parallelism)

Equipment Model (3) – Flow Line Models



- Have been used for optimization and simulation modeling studies ([5] – [8])
- Series of sequential processes $P_1, \ldots, P_M$
- Buffers modeled as zero process time modules
- Fundamental assumption: CPT is process-bound
- Modifications for CPT modeling
  - Consider robotic workload in process times of modules
  - Consider setups – reticle setup, pre-scan setup
  - Different number of processes for different lot classes
- Two types of flow lines
  - Parametric flow line (PFL) : Known process times
  - Empirical flow line (EFL) : Unknown process times

Equipment Model (3) – Flow Line Models

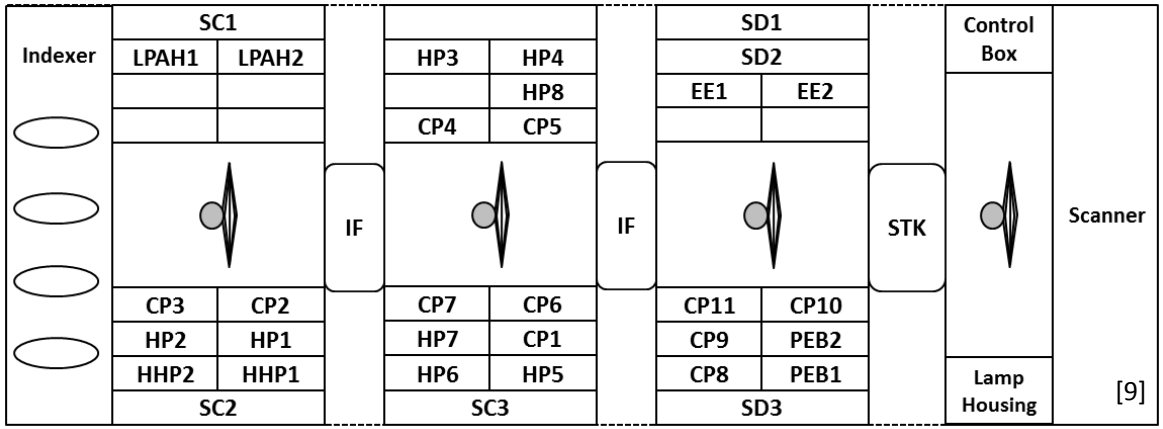
- **Notation**
  - a_w : Arrival time of wafer w to the tool, $a_w \leq a_{w+1}$
  - $X_{w,m}$: Entry time of wafer *w* into process *m* of the tool
  - $R(k,m)$: number of identical servers for process *m* for wafer class *k*
  - τ_m^k : Deterministic process time for process *m* for wafer class *k*
- **Modified Process Times**

Parametric FL	Empirical FL
$S(k,m) = \begin{cases} \tau_m^k + 3\delta + 4\varepsilon, & m = PB \\ \tau_B^k + 2\delta + 4\varepsilon, & m = B \\ \tau_B^k + \delta + 2\varepsilon, & otherwise \end{cases}$	$S(k,m) = \begin{cases} \frac{\sum_{l \in L(k)} \sum_{w=2}^{W_l} (X_{w,B+1} - X_{w-1,B+1})}{\sum_{l \in L(k)} (W_l - 1)}, & m = B \\ \min_{w,k(w)=k} (C_w - X_{w,m}), & m = M \\ \min_{w,k(w)=k} (X_{w,m+1} - X_{w,m}), & otherwise \end{cases}$

- **Elementary Evolution Equations**
  - $\tilde{X}_{w,1} = max\{a_w, \tilde{X}_{w-R'(k,1),P(w)+1}, \tilde{X}_{w-1,1}\} + \tau_s{}'(w,m)$
  - $\tilde{X}_{w,m} = max\{\tilde{X}_{w,m-1} + S(k,m-1) + \tau_R{}'(w,m), \tilde{X}_{w-R'(k,m),m+1}, \tilde{X}_{w-1,m}\}$
  - $\tilde{X}_{w,M} = max\{\tilde{X}_{w,M-1} + S(k,M-1), \tilde{X}_{w-R'(k,M),M} + S(k,M), \tilde{X}_{w-1,M}\}$
- **Start and Completion Times**
  - $\tilde{S}(i) = \tilde{X}_{\Omega(i,1),d(k)}$
  - $\tilde{C}(i) = \tilde{X}_{\Omega(i,W(i)),M} + S(k,M)$

Numerical Experiments

Numerical Experiments (1) – Detailed Model



Process order	TARC1 Process	PT	TARC2 Process	PT	BARC Process	PT
Op 1	INDEXER		INDEXER		INDEXER	
Op 2	HHP1/HHP2	80	HHP1/HHP2	80	HHP1/HHP2	80
Op 3	LPAH1/LPAH2	90	LPAH1/LPAH2	90	LPAH1/LPAH2	90
Op 4	CP2/CP3	60	CP2/CP3	60	CP2/CP3	60
Op 5	SC1/SC2	65	SC1/SC2	65	SC3	50
Op 6	SC3	50	HP1/HP2	90	HP5/HP8	90
Op 7	HP5/HP6	90	CP6/CP7	60	CP1/CP6	60
Op 8	CP10/CP11	60	SC3	50	SC1/SC2	65
Op 9	STEPPER	100	STEPPER	100	HP6/HP7	90
Op 10	PEB1/PEB2	90	PEB1/PEB2	90	CP10/CP11	60
Op 11	CP8/CP9	60	CP8/CP9	60	STEPPER	100
Op 12	EE1/EE2	90	EE1/EE2	90	PEB1/PEB2	90
Op 13	SD1/SD2/SD3	130	SD1/SD2/SD3	130	CP8/CP9	60
Op 14	HP3/HP4	90	HP3/HP4	90	EE1/EE2	90
Op 15	CP4/CP5	60	CP4/CP5	60	SD1/SD2/SD3	130
Op 16	INDEXER		INDEXER		HP3/HP4	90
Op 17					CP4/CP5	60
Op 18					INDEXER	

- Use CPT data from industry[9]; create detailed CPT model using discrete event simulation
- Two interface buffers (IF), one pre-scan buffer (STK)
- Longest waiting pair (LWP) robot policy[10]: gives optimal steady state throughput
- Robot move time : 3s, pick/place time : 1s
- Deadlock avoidance rule
- Reticle alignment setup (for every lot) ~Unif[240, 420]
- Pre-scan track setup (for lot class change) ~ Unif[210, 260]
- 15,000 lots × 30 replications
- Detailed model assumed to be exact

Numerical Experiments (2) – Simulation Description

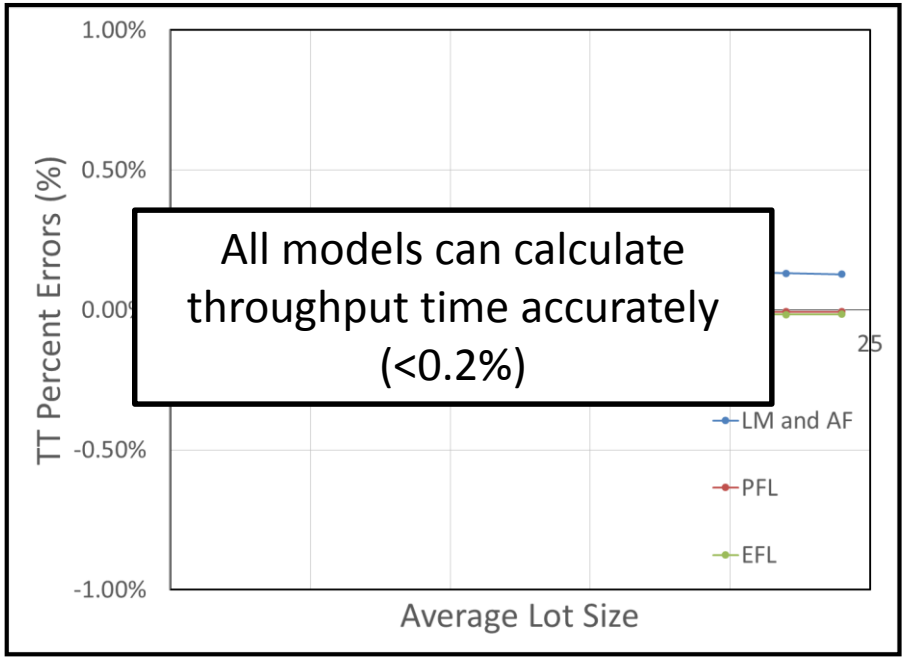
- **Type I simulation: Compare accuracy and computation**
  - Models trained on one set of input data
  - Simulated on same set of data
- **Type II simulation : Predict performance at current operating conditions**
  - Models trained on one set of input data
  - Simulated on different set of data with same parameters
- **Type III simulation : Robustness to changed operating conditions**
  - Models trained on one set of input data
  - Simulated on different set of data with different parameters

Common practice to increase tool throughput

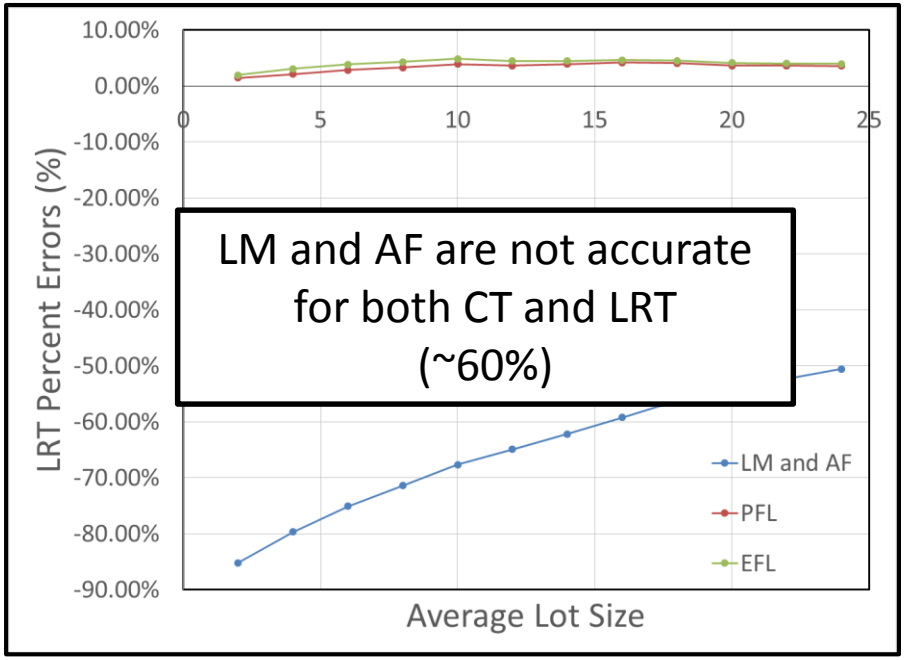
- Vary operating parameters: train size, lot size, loading level, pre-scan buffer size, penultimate bottleneck's process times, pre-scan module process times, etc.
- LM - Linear Model, AF – Affine Model, PFL – Parametric flow line, EFL – Empirical flow line, DS – Detailed Model

Numerical Experiments (3) – Type I Results

Type I Simulations



All models can calculate throughput time accurately (<0.2%)



LM and AF are not accurate for both CT and LRT (~60%)

- LM and AF are trained on throughput but cannot handle parallelism
- FL models are accurate on all metrics: CT, LRT, and TT

Numerical Experiments (3) – Type I Results

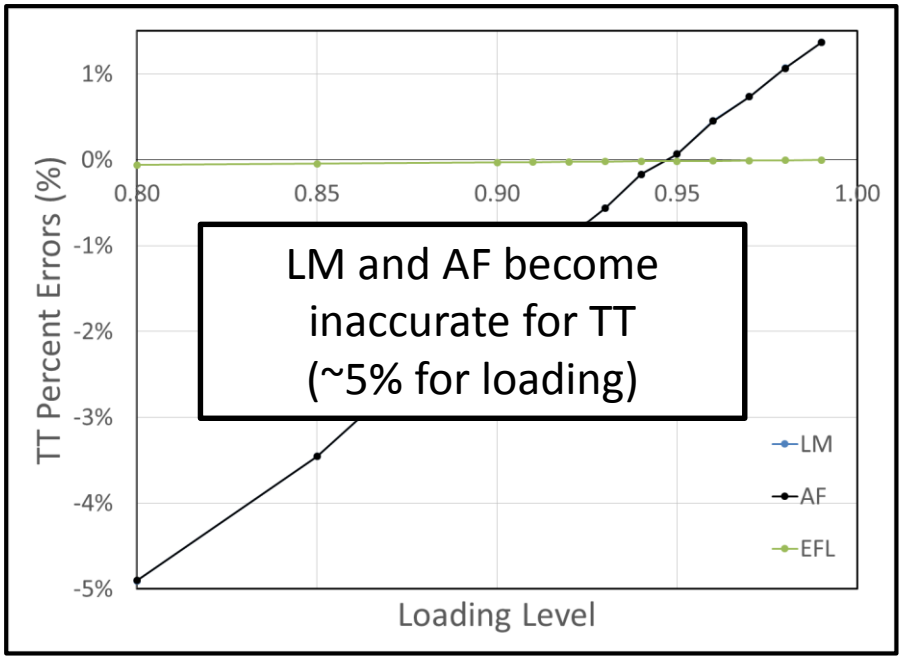
Model	Computation Time (ms)	Scaled Computation Time
LM	143	1
AF	177	1.24
PFL	83183	581
EFL	83596	584
DS	17372368	121273

- Tradeoff between accuracy and computational complexity
  - FL models approximately 500 times more complex than LM or AF
  - FL models still 200 times less computationally complex than DS
- Type II simulations show similar results to Type I
  - LM and AF correctly predict TT, not for CT or LRT

Numerical Experiments (3) – Type III Results

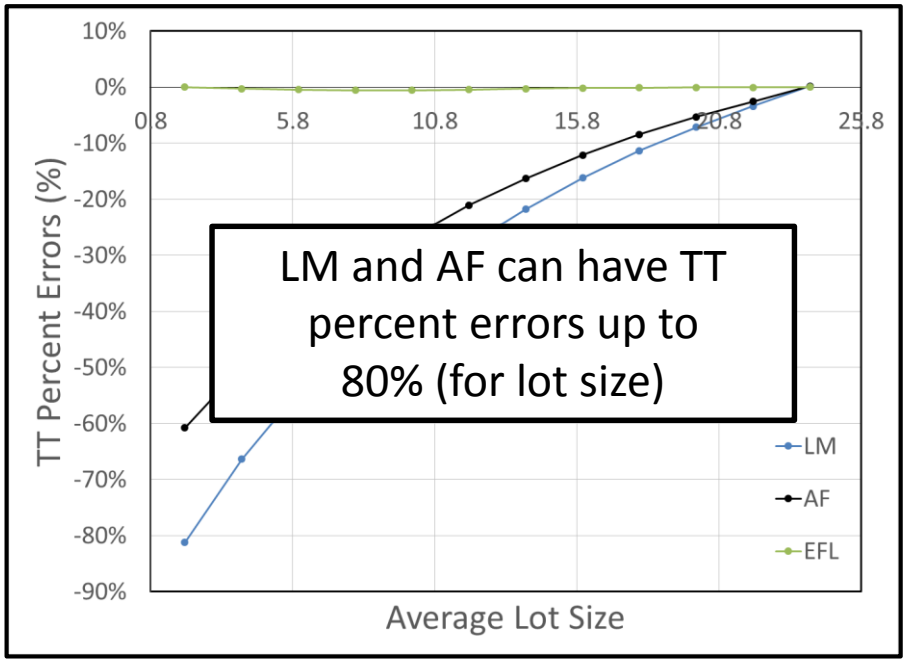
Type III Simulations

Baseline conditions: Train size = 3, lot size = {23, 24, 25}, loading = 0.95



Models trained at lot size = {23, 24, 25}

Simulated at different lot sizes



Models trained at loading = 0.95

Simulated at different loading levels

- LM and AF cannot accurately predict TT anymore; inaccurate on all metrics
- FL models are accurate on all metrics again -> robust

Numerical Experiments (3) – Accuracy Comparison

	Type I			Type II			Type III		
Linear Model	CT	LRT	TT	CT	LRT	TT	CT	LRT	TT
Affine Model	CT	LRT	TT	CT	LRT	TT	CT	LRT	TT
Flow Line Models	CT	LRT	TT	CT	LRT	TT	CT	LRT	TT

- Errors relative to detailed model
  - Error of 20%+
  - Error 5-20%
  - Error 0-5%

Concluding Remarks

Concluding Remarks

- CPT: Expensive & typically fab bottleneck

- Equipment models for CPT
  - Linear, affine, flow line models, and detailed model
  - Extension of affine model
  - Propose new method to compute module processing times for flow line
  - Assess models' fidelity on cycle time, lot residency time, and throughput time
  - Robustness to changed operating conditions

- Future work
  - Improved models: Newer exit recursions, additional parameters
  - Implementation: Fab simulation, process optimization, etc.

References

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